Beyond GICS: Business-Description Embeddings as Company Comparables

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Abstract

Choosing a company's peers is a routine step in valuation, pairs trading, index construction and risk modeling. The default peer map is GICS, which assigns each firm to one sector based on its main source of revenue. I test a text-based alternative: embed each S&P 500 firm's business description with a sentence-embedding model and treat its ten nearest neighbors as its peers. Across 499 companies, 66.0% of nearest neighbors share the firm's GICS sector, against 11.0% expected by chance (6.0×). Embedding peers also move together more than sector peers do: the mean daily-return correlation with embedding peers is 0.335, against 0.275 with same-sector firms (difference +0.060, 95% CI 0.050 to 0.069, 493 firms). Firms whose neighbors mostly sit in another sector show no such advantage, so the embeddings sharpen peer sets inside the GICS frame but do not show that GICS misplaces particular firms.

1. Introduction

Many tasks in equity research start from a peer set. A relative-value trade needs a comparable to trade against. A multiples-based valuation prices a firm off its comps. A sector fund is a peer basket by construction, and risk models use sector exposure as a control. An error in the peer set carries into each of these.

The usual peer map is the Global Industry Classification Standard. MSCI and S&P assign each firm to one of eleven sectors, and then to finer industry groups, mainly by its largest source of revenue and earnings. GICS is stable and widely used. It is also a single label set by committee, so a firm with several businesses gets one box, and a label can lag a firm whose operations have changed.

A firm's own business description offers a second view. If two firms describe their operations in similar language, they plausibly do similar things, whatever their revenue mix.

I ask two questions. First, do embedding neighbors recover GICS sector structure, and where do they depart from it? Second, do embedding peers co-move in returns more than GICS sector peers, which is the test that matters for most uses?

I set four hypotheses before the analysis:

- **H1.** Nearest neighbors mostly stay in the firm's sector, and the cross-sector links are concentrated, not uniform.
- **H1a.** Sector cohesion varies, and must be judged against a size-based baseline.
- **H1b.** Some firms are placed by GICS away from their true peers.
- **H1c.** Cross-sector links run from small, mixed sectors into large, cohesive ones.

H1 and H1a hold. H1b and H1c do not.

2. Related work

Hoberg and Phillips (2010, 2016) measure product similarity as the cosine between word vectors drawn from each firm's full 10-K business description, recomputed annually across all public firms to build time-varying competitor networks, which they distribute through the Hoberg-Phillips Data Library (Hoberg & Phillips, n.d.). This study instead applies a pretrained sentence-embedding model to the short yfinance summaries of S&P 500 firms at a single point in time, and uses the resulting neighbors to test GICS sectors and return co-movement rather than to build a replacement classification.

Bhojraj, Lee and Oler (2003) compare GICS with SIC, NAICS and the Fama-French industry definitions, and find that GICS is significantly better at explaining stock return co-movement and cross-sectional variation in valuation multiples, growth rates, R&D spending and key financial ratios.

3. Methods

3.1 Data

The universe and sector labels come from the Wikipedia list of S&P 500 constituents, taken in June 2026. Business text is the longBusinessSummary field from yfinance. The list has 502 usable listings. Three companies appear twice through dual share classes (Alphabet, Fox, News Corp) with identical summaries, so I merged each pair, leaving 499 companies in eleven sectors.

The summary is a short paragraph, not the full business section of a 10-K. The embedding therefore sees a compressed view of each firm.

3.2 Embedding

I embed each summary with all-MiniLM-L6-v2 (Reimers & Gurevych, 2019; Sentence-Transformers, n.d.), a 384-dimension sentence encoder that runs locally. The model reads at most 256 tokens, and 58.5% of summaries are longer. To avoid dropping the end of those summaries, I pack sentences into chunks of at most 254 tokens, embed each chunk, average the chunk vectors weighted by token count, and normalize the result. For summaries that fit in one chunk this gives the same vector as before.

3.3 Nearest neighbors

For each firm I take the k = 10 nearest other firms by cosine similarity, computed exactly with NumPy. As a check, I loaded the same vectors into a Qdrant vector store and confirmed that it returns identical ordered neighbor lists for all 499 firms.

3.4 Baseline and significance

If neighbors were drawn at random, the chance that a neighbor shares the firm's sector is the share of other firms in that sector. Pooled over firms, that is 11.0%. For sector i pointing to sector j, the expected share is $(n_j - [i = j]) / (N - 1)$, and a cell's lift is its observed share divided by that expectation. A lift above 1 means more links than chance and below 1 means fewer.

To test each cell I hold the neighbor graph fixed and permute sector labels across firms 10,000 times, recomputing the sector-by-sector table each time. The p-value is two-sided and measured against the mean of the permuted values. I apply the Benjamini–Hochberg correction (Benjamini & Hochberg,

1995) across the 110 off-diagonal cells. To test direction, I compare the count of links from i to j with the count from j to i for each of the 55 sector pairs, against the same permutations, again with Benjamini–Hochberg.

3.5 Mismatch candidates

A firm is a mismatch candidate when at least 70% of its neighbors are outside its GICS sector and its most common neighbor sector is not its own. The threshold is a choice. I report counts at 0.6 to 0.9 and show that the conclusion does not depend on it.

3.6 Return co-movement

I use daily adjusted closes from yfinance for 2025-07-01 to 2026-06-30, the year ending at the membership snapshot. Six firms lack enough price history, leaving 493. For each firm I compute its mean return correlation with its ten embedding peers and subtract its mean correlation with every other firm in its GICS sector. I test the mean difference with a sign-flip permutation test (10,000 draws) and a bootstrap over firms. Raw returns are the primary measure, fixed before I looked at results. Returns net of the market are the secondary measure.

4. Results

4.1 The embeddings recover sector structure

Across 499 firms, 66.0% of nearest neighbors are in the firm’s own GICS sector, against 11.0% by chance, a lift of 6.0× ($p < 0.0001$). Figure 1 shows the same thing visually: a UMAP projection of the embeddings forms sector neighborhoods. UMAP preserves only local structure, so I use it to show that clusters form and not to read distances.

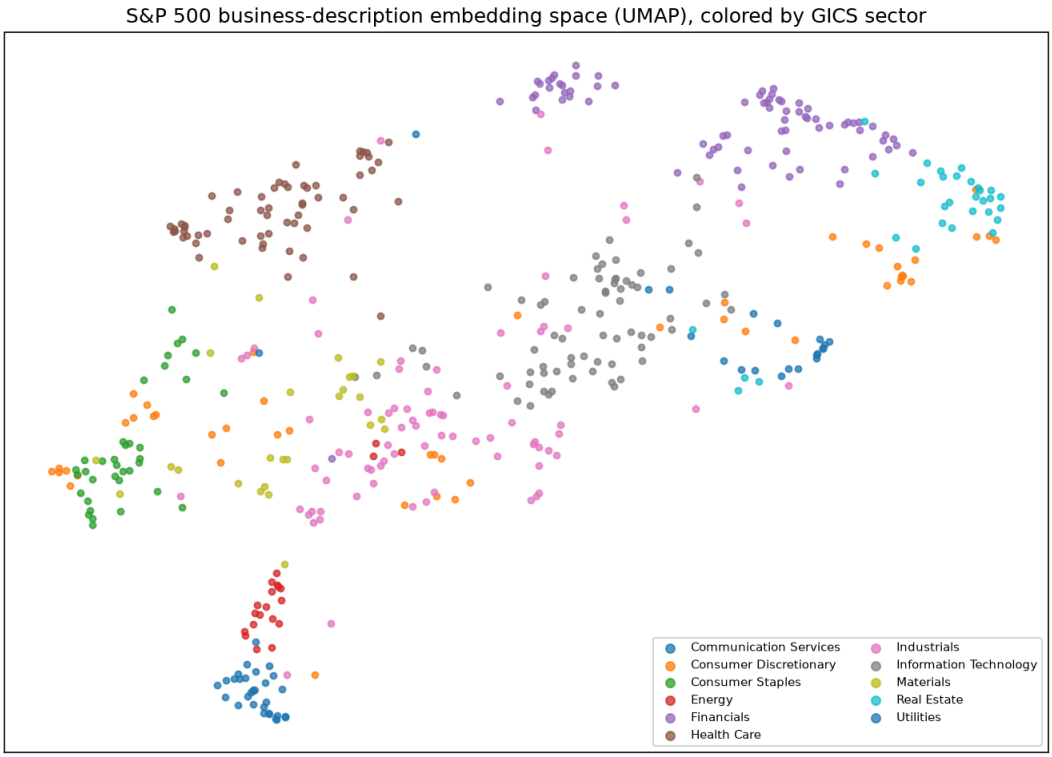


Figure 1. *UMAP projection of the 499 S&P 500 business-description embeddings (all-MiniLM-L6-v2, 384-dim), colored by GICS sector. Sectors form visible neighborhoods; UMAP is non-metric, so only local structure is interpretable.*

4.2 Cross-sector links are concentrated

The remaining 34.0% of links are not spread evenly. After correction, 91 of the 110 off-diagonal cells differ from chance. Of those, 85 are below chance, which is the expected result when most links stay in-sector. Six are above chance:

Table 1. *Sector pairs above chance after Benjamini–Hochberg correction.*

From → to	Lift
Energy → Utilities	**3.7×**
Consumer Discretionary → Consumer Staples	**2.8×**
Materials → Consumer Staples	**2.6×**
Consumer Discretionary → Real Estate	**1.8×**
Communication Services → Information Technology	**1.5×**
Materials → Industrials	**1.4×**

These are sensible overlaps: power generation and pipelines, consumer retail, chemicals and packaging, and the 2018 GICS change that formed Communication Services from telecom, media and internet names.

Eight of the 55 sector pairs are significantly one-directional. The clearest is Materials and Industrials. Materials firms have Industrials neighbors more often than chance (1.39×, q = 0.020), while

Industrials firms have Materials neighbors less often than chance (0.55×, q = 0.035). That is 58 links one way and 23 the other (q = 0.0055).

I expected these one-directional links to run from small, mixed sectors into large, cohesive ones (H1c). They do not. Among the eight significant pairs, the receiving sector is larger in five and more cohesive in five, which is close to a coin flip. Materials → Industrials itself runs into the least cohesive sector in the index.

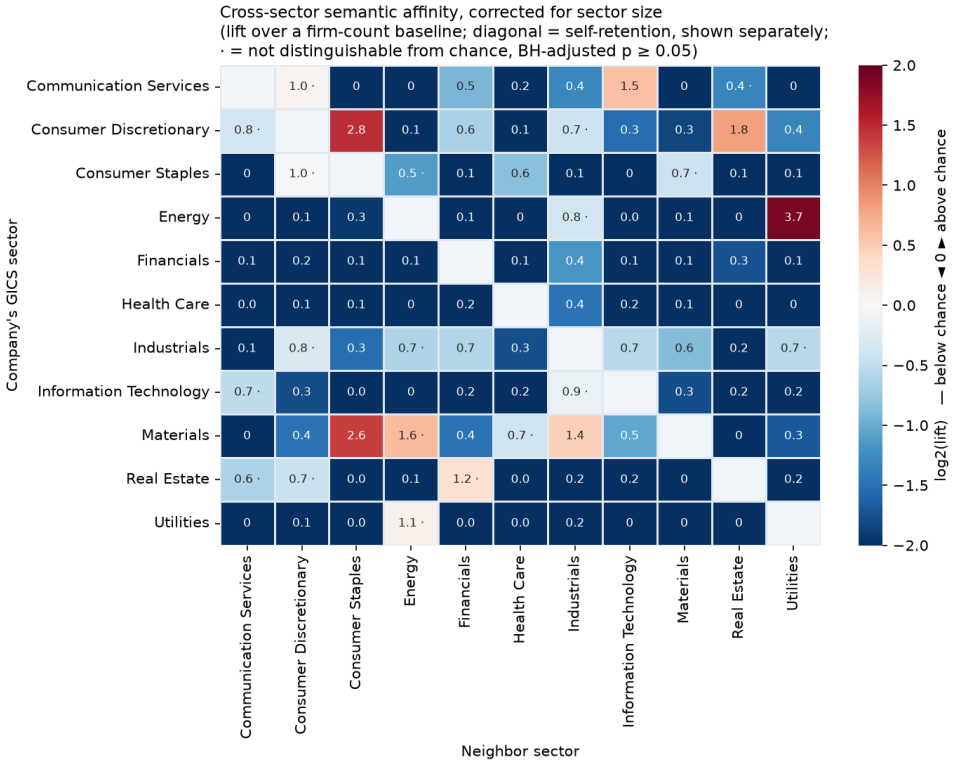


Figure 2. *Cross-sector affinity as lift over the firm-count baseline (observed share divided by expected share), on a $\log_2$ color scale centered at chance. A dot marks cells with Benjamini–Hochberg adjusted p ≥ 0.05. Self-retention is shown separately.*

4.3 Cohesion depends on adjusting for size

Raw self-retention makes large sectors look cohesive because they have more firms available as neighbors. Dividing each sector's retention by its own chance rate removes that advantage (Table 2, Figure 3). Utilities (14.9×) and Energy (14.8×) are the tightest. Industrials (3.5×) and Consumer Discretionary (3.6×) are the loosest: they are large sectors that span unrelated businesses. This supports H1a.

Table 2. *Sector cohesion, raw and size-adjusted, sorted by lift. Self-retention is observed intra-sector neighbor rate; chance is the per-sector random baseline (n− 1)/(N− 1); lift is their ratio.*

Sector	n	Self-retention	Cross-rate	Chance	Lift
Utilities	31	90.0%	10.0%	6.0%	**14.9×**
Energy	21	59.5%	40.5%	4.0%	**14.8×**
Communication Services	20	48.5%	51.5%	3.8%	**12.7×**
Real Estate	31	64.5%	35.5%	6.0%	**10.7×**
Consumer Staples	36	74.2%	25.8%	7.0%	**10.6×**
Health Care	59	86.6%	13.4%	11.6%	**7.4×**
Financials	75	84.7%	15.3%	14.9%	**5.7×**
Materials	26	25.8%	74.2%	5.0%	**5.1×**
Information Technology	72	70.6%	29.4%	14.3%	**4.9×**
Consumer Discretionary	48	34.0%	66.0%	9.4%	**3.6×**
Industrials	80	55.0%	45.0%	15.9%	**3.5×**

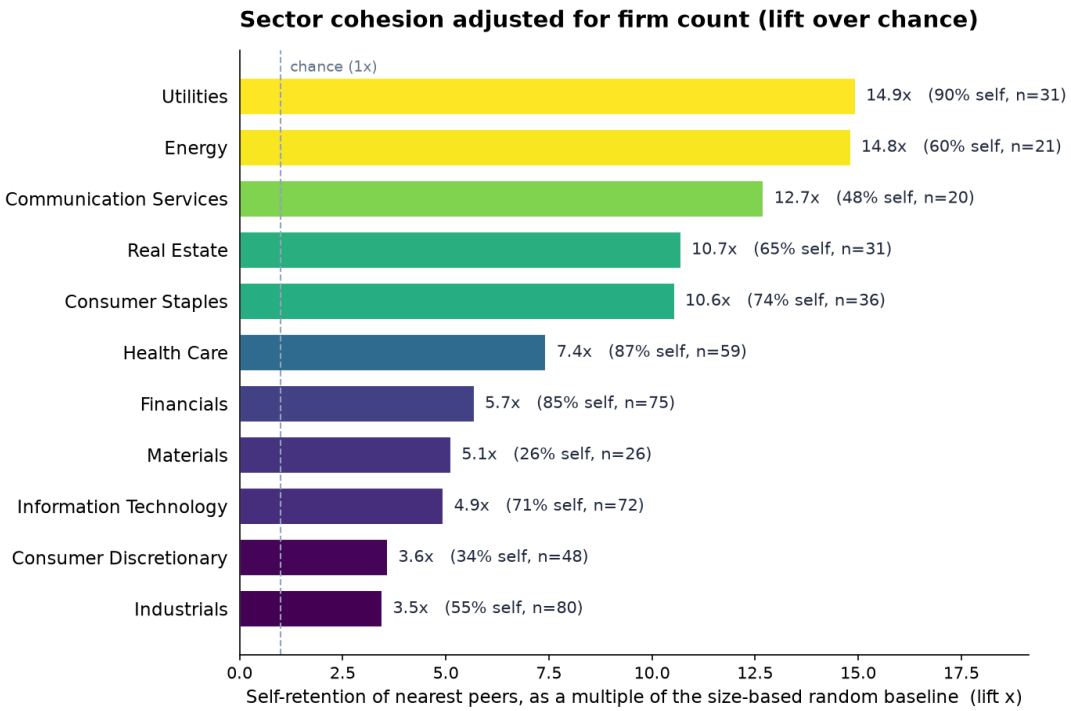


Figure 3. *Sector cohesion as a multiple of the size-based random baseline (lift over chance). Energy and Utilities are tightest once size is accounted for; Industrials and Consumer Discretionary are loosest, despite middling raw self-retention.*

4.4 Embedding peers co-move more than sector peers

This is the main result. A firm’s mean daily-return correlation with its ten embedding peers is 0.335. Its mean correlation with firms in its GICS sector is 0.275. The difference is +0.060 (95% CI 0.050

to 0.069, $p < 0.0001$), and embedding peers are the more correlated set for 72% of the 493 firms. Net of the market, the difference is +0.070 (95% CI 0.060 to 0.080).

Two thirds of embedding peers are in the firm's own sector, so this result mostly reflects the embeddings picking the closest firms within a sector. I did not compare against GICS sub-industry peers, which would be the stricter benchmark.

4.5 Mismatch candidates show no return advantage

At the 0.7 threshold, 83 firms are mismatch candidates. If GICS had misplaced these firms, their embedding peers should co-move with them more than their sector peers do. They do not: the difference is −0.014 (95% CI −0.036 to 0.010, $p = 0.25$). The result is null at every threshold from 0.6 to 0.9 (98, 83, 57 and 27 candidates). H1b is not supported.

Table 3 lists four candidates; Figure 4 shows the neighbors for three. Broadridge is classed as Industrials, and its nearest neighbors are four Financials firms (Northern Trust, Bank of America, State Street, Raymond James) and IBM (Information Technology). Berkshire Hathaway is classed as Financials, and its neighbors span Consumer Discretionary (Genuine Parts, Tractor Supply) and Industrials (Wabtec, Norfolk Southern, Eaton), which fits a conglomerate with no single peer group. Ecolab (Materials) has two Health Care life-sciences tools firms (Waters, Agilent) among its five nearest. Aptiv (Consumer Discretionary) has two Information Technology firms and Rockwell Automation among its five nearest, at cosines of 0.45 to 0.50. These firms describe themselves unlike their sector. The return data does not show that they trade unlike it.

Table 3. *Case-study firms flagged as mismatches (cross-sector rate ≥ 0.7 with a foreign modal neighbor sector). Cosine ranges are for each firm's five nearest neighbors; Figure 4 shows them for the first three.*

Firm	GICS → Semantic	Neighbor cosine	Read
Berkshire Hathaway (BRK.B)	Financials → Industrials	0.54–0.58	Neighbors are Consumer Discretionary (Genuine Parts, Tractor Supply) and Industrials (Wabtec, Norfolk Southern, Eaton). Consistent with a conglomerate that has no single peer group.
Broadridge (BR)	Industrials → Financials	0.61–0.62	Four Financials firms (Northern Trust, Bank of America, State Street, Raymond James) and IBM. It describes itself like financial infrastructure.
Ecolab (ECL)	Materials → Health Care	0.45–0.55	Two of its five nearest are Health Care life-sciences tools firms (Waters, Agilent).
Aptiv (APTV)	Consumer Discretionary → Information Technology	0.45–0.50	Auto-components on paper; the description is ADAS and software-defined-vehicle technology.

BRK.B (GICS: Financials) — nearest semantic neighbors

Ticker	Name	Sector	Cos
GPC	Genuine Parts Company	Consumer Discretionary	0.58
WAB	Wabtec	Industrials	0.57
TSCO	Tractor Supply	Consumer Discretionary	0.56
NSC	Norfolk Southern	Industrials	0.56
ETN	Eaton Corporation	Industrials	0.54

BR (GICS: Industrials) — nearest semantic neighbors

Ticker	Name	Sector	Cos
NTRS	Northern Trust	Financials	0.62
BAC	Bank of America	Financials	0.62
IBM	IBM	Information Technology	0.61
STT	State Street Corporation	Financials	0.61
RJF	Raymond James Financial	Financials	0.61

ECL (GICS: Materials) — nearest semantic neighbors

Ticker	Name	Sector	Cos
CLX	Clorox	Consumer Staples	0.55
WAT	Waters Corporation	Health Care	0.46
A	Agilent Technologies	Health Care	0.46
IFF	International Flavors & Fragrances	Materials	0.46
BAC	Bank of America	Financials	0.45

Figure 4. *Five nearest neighbors, with cosine similarity, for Broadridge, Berkshire Hathaway and Ecolab.*

5. Limitations

- The text is a short summary, not a full filing.
- The model is general-purpose, with no finance tuning.
- The analysis is one snapshot of membership and one year of returns.
- The peers come from June 2026 descriptions and the return window ends in June 2026, so the co-movement test is in-sample, not a forecast.
- The co-movement tests treat firms as independent, although peer sets overlap, so the confidence intervals are somewhat narrow.
- The benchmark is the GICS sector, not the sub-industry.
- k = 10 and the 0.7 threshold are choices.

6. Conclusion

Business-description embeddings recover GICS sectors at six times the chance rate, and the peers they select co-move more than sector peers do. The practical use is a finer peer set: ten named comparables per firm that track it more closely than its sector average. The data does not support the stronger claim that embeddings reveal firms GICS has put in the wrong sector.

Three extensions follow directly: compare against GICS sub-industry peers, use full 10-K text and a finance-tuned model, and repeat the snapshot over several years to see whether the co-movement advantage persists.

Revision note

An earlier version of this paper used truncated summaries and a different permutation test. It reported a pattern of links from small sectors into large ones and described several firms as misplaced by GICS. Neither claim holds under the corrected analysis reported here.

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